Psoriasis and suicidality: A systematic review and meta-analysis

Sanminder Singh, BS, BA,[a] Catherine Taylor, BS,[b] Heather Kornmehl, BS,[c] and April W. Armstrong, MD, MPH[d,e]

Davis, Los Angeles, and Sacramento, California, and Philadelphia, Pennsylvania

Background: Psoriasis is associated with psychiatric comorbidities; however, the relationship between psoriasis and suicidality is not well understood.

Objective: To perform a systematic review and meta-analysis that elucidates the relationship between psoriasis and suicidality.

Methods: Applying the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines, we systematically searched the PubMed, EMBASE, PsycINFO, and Cochrane databases. We searched literature published between 1946 and 2017.

Results: We identified 18 studies with a total of 1,767,583 participants, of whom 330,207 had psoriasis. On the basis of random effects modeling, the pooled odds ratio (OR) for suicidal ideation among patients with psoriasis was 2.05 (95% confidence interval [CI], 1.54-2.74). Patients with psoriasis were more likely to exhibit suicidal behaviors (combined attempted and completed suicides) with a pooled OR of 1.26 (95% CI, 1.13-1.40). Subgroup analysis showed that patients with psoriasis were more likely to attempt suicides (OR, 1.32; 95% CI, 1.14-1.54) and complete suicide (OR, 1.20; 95% CI, 1.04-1.39) than those without psoriasis. More severe psoriasis and younger age were associated with greater likelihood of suicidality.

Limitations: There are few studies examining suicidality in conjunction with psoriasis severity.

Conclusions: Patients with psoriasis have a significantly higher likelihood of suicidal ideation, suicide attempts, and completed suicides. Among patients with psoriasis, those who are younger and whose psoriasis is more severe are at particular risk for suicidality. (J Am Acad Dermatol 2017;77:425-40.)

Key words: completed suicide; meta-analysis; prevalence; psoriasis; suicidal ideation; suicidality; suicide attempts; systematic review.

Psoriasis is a chronic, inflammatory skin disorder that affects 2% to 4% of the general population, with recent estimates suggesting more than 125 million affected individuals worldwide.[1,2] Psoriasis is characterized by pruritic and painful lesions that can significantly affect patient quality of life, even when the disease is not extensive.[3] Additionally, psoriasis has been associated with extracutaneous comorbidities such as inflammatory arthritis and cardiometabolic disorders,[4,5] which also add to the overall burden of disease.

Abbreviations used:

CI:	confidence interval
HR:	hazard ratio
IL:	interleukin
OR:	odds ratio

From the University of California Davis School of Medicine, Sacramento[a]; University of California Davis[b]; Drexel University College of Medicine, Philadelphia[c]; and the Southern California Clinical and Translational Science Institute[d] and Keck School of Medicine,[e] University of Southern California, Los Angeles.

Funding sources: None.

Disclosure: Dr Armstrong has served as an investigator and consultant/advisor to AbbVie, Amgen, Celgene, Janssen, Lilly, Novartis, Regeneron, Pfizer, and Modernizing Medicine. Drs Singh, Taylor, and Kornmehl have no conflicts of interest to declare.

Accepted for publication May 19, 2017.

Reprints not available from the authors.

Correspondence to: April W. Armstrong, MD, MPH, University of Southern California, 1975 Zonal Ave, KAM 510, MC 9034, Los Angeles, CA 90089. E-mail: aprilarmstrong@post.harvard.edu.

0190-9622/$36.00

© 2017 Published by Elsevier on behalf of the American Academy of Dermatology, Inc.

http://dx.doi.org/10.1016/j.jaad.2017.05.019

Studies have reported that up to half of patients with psoriasis experienced feelings of profound helplessness and isolation on account of the functional limitations and social and psychological impact of their disease.[6,7] Because of its pervasive nature, psoriasis has been thought to contribute to the development and progression of psychiatric disorders. One study found that patients with psoriasis are 1.5 times more likely to have depression and more than 4 times as likely to use antidepressants as controls without psoriasis.[8] Patients with psoriasis are also at higher risk of receiving a diagnosis of anxiety disorder than the general population.[9,10]

There is a gap in the literature examining the association between psoriasis and suicidality. Population-based studies suggest that patients with psoriasis may have an increased prevalence of suicidality. Suicidality may be defined as suicidal ideation, suicide attempts, or completed suicides. Of these 3 aspects, suicide attempts and completed suicides are collectively considered suicidal behavior. In this systematic review and meta-analysis, we aim to synthesize the available literature on the epidemiologic association between psoriasis and the various aspects of suicidality.

CAPSULE SUMMARY

- Some studies suggest a relationship between psoriasis and suicidality.
- This meta-analysis provides further evidence that patients with psoriasis have a higher likelihood of suicidality than those without psoriasis.
- These findings emphasize the importance of inquiring about prior or current suicidality among patients with psoriasis and making appropriate referrals.

METHODS

To examine the epidemiologic association between psoriasis and suicidality, we followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines to perform a systematic review search using the PubMed, EMBASE, PsycINFO, and Cochrane databases. The review protocol was registered with PROSPERO (CRD42017054446). We used the following search criteria: (*Psoriasis* [MeSH]) AND (*Suicide* [MeSH] OR *Suicide, Attempted* [MeSH] OR *suicid**). We used the term *suicid** to include other variations of the word such as suicidal or suicidality. The search was limited to English-language studies and included studies published from database inception to the present, 1946–2017. The initial search yielded 261 references (Fig 1).

Two authors (S.S. and C.T.) read reference abstracts to determine eligibility for inclusion in the systematic review. We applied inclusion/exclusion criteria to 200 unique references. For this meta-analysis, we focused on noninterventional studies (eg, case-control, cross-sectional, and cohort studies) to examine the prevalence of suicidality rather than change in suicidality in response to interventions. The articles must have also fulfilled all of the following inclusion criteria: the article must be written in English; the study participants must be adults (≥18 years); the diagnosis of psoriasis must be documented; suicidality must be a primary or secondary end point, and it must be documented through use of a questionnaire/survey or patient medical records; and suicidality must be assessed in conjunction with psoriasis and numerically reported (numbers of patients affected, percent values, odds ratio [OR] or hazard ratio [HR]). Case reports, case series, review articles, letters, and commentaries were excluded. We identified 75 articles to be reviewed in full. In the process of screening articles for inclusion, we identified 2 additional articles for inclusion by manually searching article references.

S.S. and C.T. independently reviewed 75 full-text articles for eligibility and excluded articles with the following characteristics because they did not fulfill the inclusion criteria for the systematic review: no report of suicidality in conjunction with psoriasis (n = 7), full-length article unavailable (n = 5), case report/series/review article (n = 21), suicidality not evaluated or prevalence not reported (n = 12), psoriatic arthritis only (n = 1), nonobservational studies (n = 9), and included pediatric patients (n = 2). After these exclusions, 18 unique articles were eligible for inclusion in our systematic review.

During data extraction for this systematic review, discrepancies were resolved by consensus among S.S., C.T., H.K., and A.W.A. For each study, we recorded the study year; study setting; patient demographic characteristics (age and sex); prevalence of suicidal ideation, attempts, or completion; and diagnoses of depression and anxiety where available. To measure study quality, we used a validated assessment tool for observational studies.[11,12]

Of the 18 included studies, 13 examined at least 1 aspect of suicidality (ideation, attempts, or completed) in patients with psoriasis versus in controls without psoriasis. Eight of those 12 studies specifically reported suicidal ideation as percent values or numbers of patients experiencing suicidal

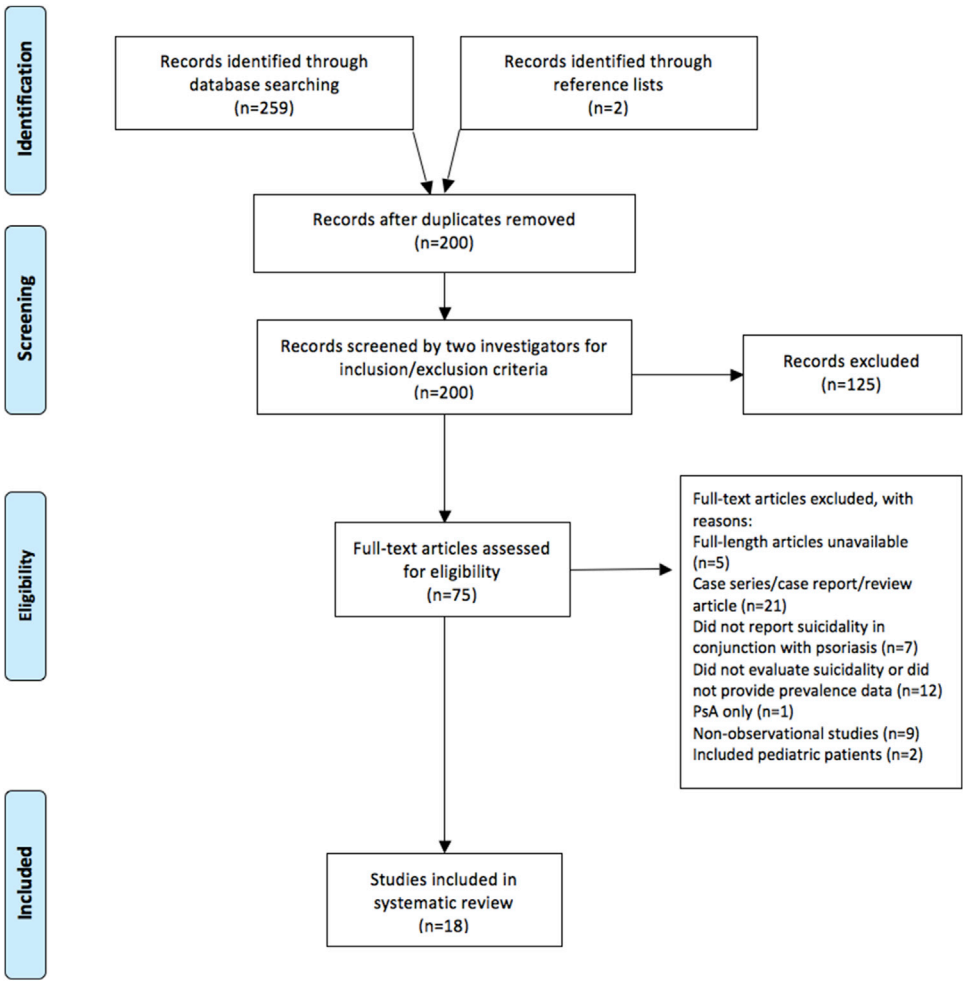


Fig 1. Selection process for study inclusion in the systematic review. *PsA*, Psoriatic arthritis.

ideation in the psoriasis group versus in the control group.

To synthesize the results of those studies into a meta-analysis, we calculated unadjusted ORs with STATA software, version 14.1 (Stata Corp LP, College Station, TX).[13] Using OR data obtained from the published results, we estimated the pooled OR for the presence of suicidal ideation and the pooled OR for the prevalence of patients with psoriasis who acted on their suicidal ideation. Additionally, we performed sensitivity analysis to separately examine prevalence of suicide attempts and completed suicides in patients with psoriasis versus in controls. We used the random effects analysis model of Dersimonian and Laird to generate all pooled ORs and account for any study heterogeneity.[14]

Publication bias may arise in this field of investigation as a result of studies with statistically significant results being more likely to be published and cited. We assessed publication bias by using visual inspection of a funnel plot representing the estimate of effect from each study in the meta-analysis against a measure of its precision. The Egger regression test was done to calculate a bias coefficient and confirm the findings of the funnel plot. The findings of the systematic review and meta-analysis are included in the results section.

All analyses were performed using STATA software, version 14.1.

RESULTS

In this systematic review, we searched literature spanning from 1946 to 2017 and synthesized data on 1,767,583 study participants, of whom 330,207 had psoriasis (Table I[15-32]). Every study included in this systematic review examined at least 1 of the following aspects of suicidality: suicidal ideation, suicide attempts, and completed suicides. Because 2 of the studies did not report separate numbers for suicidal ideation, suicide attempts, or completed suicide, they could not be included in

Table I. Study population characteristics: psoriasis and suicidality

Study	Study setting	Study design (follow-up)	No. of patients with psoriasis	No. of control participants	Mean age of patients with psoriasis, y (±SD)*	Mean age of controls, y (±SD)*
Gupta et al[15] (1993)	United States; outpatient (dermatology department)	Cross-sectional	217	0	48 (±16)	NR
Rapp et al[16] (1997)	United States; outpatient (dermatology department and clinics)	Cross-sectional	297	0	49	NR
Gupta and Guptal[17] (1998)†	Canada; inpatient and outpatient (hospital, clinic, and general community)	Cross-sectional	217 (79 inpatients, 138 outpatients)	72 with acne; 45 with alopecia areata; 146 with atopic dermatitis	47 (±16)	24 (±7) with acne; 45 (±12) with alopeacia areata; 42 (±16) with atopic dermatitis
Krueger et al[18] (2001)	United States; National Psoriasis Foundation participants (general community)	Cross-sectional	17,488	0	54	NR
Sharma et al[19] (2001)	India; inpatient (hospital)	Case-control	30	30 with vitiligo	NR	NR
Zachariae et al[20] (2004)	Denmark; inpatient and outpatient (hospital and dermatology department clinics)	Case-control	113 (57 inpatients and 56 outpatients)	293 general population	49.9	44.2
Friedman et al[21] (2006)	Israel; outpatient (mental health department clinic)	Case-control	1102	1419 with acne	NR	NR
Picardi et al[22] (2006)	Italy; outpatient (dermatology department)	Cross- sectional	80	386 with general skin disease other than psoriasis	NR	NR
Abuabara et al[23] (2010)	United Kingdom; outpatient (General Practice Research Database)	Cohort (15 y)	3603	14,330 general population	52	50

Kurd et al[24] (2010)	United Kingdom; outpatient (General Practice Research Database)	Cohort (15 y)	149,998 total Mild psoriasis: 146,042	766,950 total Mild psoriasis—matched general population controls: 746,930	Mild psoriasis: 40	Mild psoriasis—matched general population controls: 33
			Severe psoriasis: 3956	Severe psoriasis—matched general population controls: 146,042	Severe psoriasis: 48	Severe psoriasis—matched general population controls: 34
Sanna et al[25] (2013)	Australia; (Geelong Osteoporosis Study)	Cross- sectional	39	868 with general medical disease other than psoriasis	NR	NR
Kimball et al[26] (2014)	United States, Canada, Europe, and Latin America; (PSOLAR)	Cohort (Ongoing)	11900	0	49 (±14)	NR
Svedbom et al[27] (2015)	Sweden; (VEGA register and Skane Health Care Register)	Cohort (10 y)	39,074 total Mild psoriasis: 34,355	154,775 total Mild psoriasis—matched general population controls: 136,409	Mild psoriasis: 49 (±19)	Mild psoriasis—matched general population controls: 49 (±3)
			Severe psoriasis: 4719	Severe psoriasis—matched general population controls: 18,366	Severe psoriasis: 52 (±16)	Severe psoriasis—matched general population controls: 52 (±16)
Dalgard et al[28] (2015)	Belgium, Denmark, France, Germany, Hungary, Italy, Netherlands, Norway, Poland, Russia, Spain, Turkey, and United Kingdom; outpatient (dermatology outpatient clinics)	Cross-sectional	626	1359 general population	NR	41 (±14)
Egeberg et al[29] (2016)	Denmark (Danish National Patient Register and National Causes of Death Registry)	Cohort (14 y)	68,511 total Mild psoriasis: 57,502	340,152 total Mild psoriasis- matched general population controls: 285,544	Mild psoriasis: 52 (±17)	Mild psoriasis- matched general population controls: 52 (±17)
			Severe psoriasis: 11,009	Severe psoriasis-matched general population controls: 54,608	Severe psoriasis: 48 (±15)	Severe psoriasis-matched general population controls: 48 (±15)

Continued

Table I. Cont'd

Study	Study setting	Study design (follow-up)	No. of patients with psoriasis	No. of control participants	Mean age of patients with psoriasis, y (±SD)*	Mean age of controls, y (±SD)*
Pompili et al[30] (2016)	Italy; inpatient (dermatology department)	Cross- sectional	91	66 general skin and medical diseases other than psoriasis	51 (±15)	49 (±18)
Lamb et al[31] (2016)	England; NR (questionnaires in IMPARTS system)	Cross-sectional	607	0	47 (±14)	NR
Wu et al[32] (2017)	United States (GHC and KPSC database)	Cohort (8.5 y)	36,214	156,093	54 (±16)	54 (±16)

GHC, Group Health Cooperative; *IMPARTS*, Integrating Mental and Physical Healthcare, Research, Training, and Services; *KPSC*, Kaiser Permanente Southern California; *NR*, not reported; *PSOLAR*, Psoriasis Longitudinal Assessment and Registry; *SD*, standard deviation.
*Standard Deviation is provided when made available in the original articles.
†This study reanalyzed data from the patients with psoriasis in Gupta et al.[15]

the meta-analysis.[21,24] Table I summarizes the study and participant characteristics, and Table II[15-32] summarizes the study findings.

Suicidality was evaluated by using assessment tools such as the Beck Depression Inventory and Patient Health Questionnaire, health and government agency records, and miscellaneous surveys. Out of the 18 articles, 13 reported suicidality using percentages,[15-22,25-27,30,31] 2 reported HRs,[23,24] 2 reported incidence rate ratio,[29,32] and 1 reported an adjusted OR after adjustment for sex, age, socioeconomic status, stress, comorbidities, and depression.[28] Thirteen studies reported the prevalence of suicidality in patients with psoriasis versus in controls without psoriasis, whereas 5 studies did not have a control group without psoriasis. None of the studies scored 10 points on the modified Newcastle-Ottawa scale for quality assessment. Five studies scored a maximum of 8 out of 10 points, whereas the majority of studies scored between 5 and 7 points (n = 10).

Of the 15 studies that examined suicidal ideation in patients with psoriasis, 12 reported a higher prevalence or rate in patients with psoriasis than in controls without psoriasis. In 1 of the largest studies examining suicidal ideation, with a sample of 1985 participants, the authors adjusted for sex, age, socioeconomic status, stress, comorbidities, and depression when reporting effect size for suicidal ideation. They reported that patients with psoriasis had a significantly higher likelihood of having suicidal ideation compared with controls (adjusted OR, 1.94; 95% CI, 1.33-2.82).[28] Three out of the 15 studies examining suicidal ideation reported lower prevalence of suicidal ideation in patients with psoriasis versus in controls; however, the control groups consisted of patients with urticarial disease, severe acne, and major chronic illnesses such as pulmonary and liver disease.[21,22,25]

Of the 15 studies examining suicidal ideation, 8 assessed suicidal ideation in patients with psoriasis versus in controls without psoriasis. After synthesizing the aforementioned 8 studies on suicidal ideation, we found that patients with psoriasis were more than twice as likely to have suicidal ideation than patients without psoriasis (pooled OR, 2.05; 95% CI, 1.54-2.74) (Fig 2). The funnel plot of the 8 studies demonstrated symmetry, suggesting limited publication bias, which was confirmed by the Egger regression test (P = .359) (Supplemental Fig 1; available at http://www.jaad.org). The I^2 value of 44.2% suggested moderate heterogeneity. Of note, for 1 study that used multiple qualitative interview questions to assess suicidal ideation, we drew upon the data from statistically significant comparisons

Table II. Study outcomes: psoriasis and suicidality

Study	Outcome ascertainment	Aspects of suicidality assessed	Suicidality in psoriasis, n (%)	Suicidality in controls, n (%)	Psoriasis and suicidality: measure of effect
Gupta et al[15] (1993)	CRSD	Ideation	"Dying is the best solution for me" (answered yes): 12/215 (5.6%); "I often wish I were dead" (answered yes): 21/217 (9.7%); "I feel that life is still worth living" (answered no): 18/215 (8.4%); "I have been thinking about trying to kill myself" (answered yes): 12/217 (5.5%)	NR	NR
Rapp et al[16] (1997)	Qualitative questionnaire (Investigator designed)	Ideation and attempt	Wished for death because of psoriasis: 74/297 (25%); Believed that life was not worth living because of psoriasis: 24/297 (8%); Had tried to hurt or kill themselves because of psoriasis: 5/297 (1.6%)	NR	NR
Gupta and Gupta[17] (1998)	CRSD	Ideation	"Dying is the best solution for me" (answered yes): Total: 10/213 (4.7%), Outpatient: 4/77, Inpatient: 6/136; "I often wish I were dead" (answered yes): Total: 21/217 (9.7%), Outpatient: 7/79, Inpatient: 14/138; "I feel that life is still worth living" (answered no): Total: 18/214 (8.4%), Outpatient: 4/78, Inpatient: 14/136; "I have been thinking about trying to kill myself" (answered yes):	"Dying is the best solution for me" (answered yes): Acne 0/71 (0%); Alopecia areata 1/45 (2.2%), Atopic dermatitis 2/146 (1.4%); "I often wish I were dead" (answered yes): Acne 6/72 (8.3%), Alopecia areata 4/44 (9.1%), Atopic dermatitis 4/146 (2.7%); "I feel that life is still worth living" (answered no): Acne 1/72 (1.4%), Alopecia areata 1/45 (2.2%), Atopic dermatitis 5/146 (3.4%);	OR, 3.36 (95% CI, 1.38-8.20)*

Continued

Table II. Cont'd

Study	Outcome ascertainment	Aspects of suicidality assessed	Suicidality in psoriasis, n (%)	Suicidality in controls, n (%)	Psoriasis and suicidality: measure of effect
			Total: 12/217 (5.5%), Outpatient: 2/79, Inpatient: 10/138	"I have been thinking about trying to kill myself" (answered yes): Acne 4/72 (5.6%), Alopecia areata 0/44 (0%), Atopic dermatitis 3/143 (2.1%)	
Krueger et al[18] (2001)	Qualitative questionnaire (Investigator designed)	Ideation	Ages 18-34: 192/1918 (10%); Ages 35-54: 464/6625 (7%); Ages ≥55: 267/8891 (3%)	N/A	NR
Sharma et al[19] (2001)	GHQ-H	Ideation and attempt	Suicidal ideation: 8/30 (26.6%) Suicide attempts: 1/30 (3.3%)	Suicidal ideation: 3/30 (10%) Suicide attempts: 1/30 (3.3%)	Suicidal ideation: OR, 3.27 (95% CI, 0.77-13.82)* Suicide attempts: OR, 1.00 (95% CI, 0.06-16.76)*
Zachariae et al[20] (2004)	BDI	Ideation	24/113 (21.2%)	Healthy controls: 20/293 (6.8%); Atopic dermatitis: 18/95 (18.9%); Eczema: 7/120 (5.8%); Urticaria: 2/32 (6.3%)	OR, 3.68 (95% CI, 1.94-6.98)*
Friedman et al[21] (2006)	IDF Mental Health Department Database and General Medical Database	Ideation and attempt	39/1102 (3.5%)	106/1419 (7.5%)	NR
Picardi et al[22] (2006)	PHQ	Ideation	8/80 (10%)	Nevi, mycosis, androgenetic alopecia: 0 (0%) Seborrheic dermatitis: 1/19 (5.3%) Bacterial infections: 1/15 (6.7%) Benign skin neoplasms: 2/28 (7.1%) Acne: 3/42 (7.1%) Skin tumors: 1/12 (8.3%) Various forms of dermatitis: 5/57 (8.8%) Urticaria: 3/16 (18.8%)	OR, 1.61 (95% CI, 0.66-3.92)*

Abuabara et al[23] (2010)	General Practice Research Database	Completed	1	1	Completed suicide: OR, 3.98 (95% CI, 0.25-63.65)* HR, 3.35 (95% CI, 0.21-53.77)
Kurd et al[24] (2010)	Oxford Medical Information System, Read Codes	Ideation, attempt, and completed	Unadjusted incidence rate per 1000 person-years: Mild psoriasis: 0.93 (95% CI, 0.85-1.00) Severe psoriasis: 0.92 (95% CI, 0.57-1.41)	Unadjusted incidence rate per 1000 person-years: Mild psoriasis-matched general population control: 0.66 (95% CI, 0.63-0.68) Severe psoriasis-matched general population controls: 0.66 (95% CI, 0.52-0.82)	Adjusted for age and sex: All psoriasis: HR, 1.44 (95% CI, 1.32-1.57) Mild psoriasis: HR, 1.44 (95% CI, 1.32-1.57) Severe psoriasis: HR, 1.51 (95% CI, 0.92-2.49) Stratified by age: Age 20 y: HR, 1.83 (95% CI, 1.64-2.05) Age 40 y: HR, 1.38 (95% CI, 1.27-2.51) Age 60 y: HR, 1.04 (95% CI, 0.91-1.20)
Sanna et al[25] (2013)	Qualitative questionnaire (Investigator-designed)	Ideation	3/39 (7.7%)	74/868 (8.5%)	OR, 0.89 (95% CI, 0.27-2.96)*
Kimball et al[26] (2014)	PSOLAR database	Ideation	114/11,900 (1%)	NR	NR†
Svedbom et al[27] (2015)	CDR contains information from death certificates	Completed	Completed suicides: Mild psoriasis: 27 Severe psoriasis: 3	Completed suicides: Mild psoriasis-matched general population controls: 84 Severe psoriasis-matched general population controls: 12	OR, 1.24 (95% CI, 0.82-1.87)* Mild psoriasis: HR, 1.28 (95% CI, 0.83-1.97) Severe psoriasis: HR, 0.99 (95% CI, 0.28-3.57)

Continued

Table II. Cont'd

Study	Outcome ascertainment	Aspects of suicidality assessed	Suicidality in psoriasis, n (%)	Suicidality in controls, n (%)	Psoriasis and suicidality: measure of effect
Dalgard et al[28] (2015)	Qualitative questionnaire (Investigator designed)	Ideation	106/626 (17.3%)	88/1359 (8.3%)	AOR, 1.94 (95% CI, 1.46-2.58)
Egeberg et al[29] (2016)	Danish National Patient Register, ICD-8 and ICD-10 codes	Attempt and completed	Suicide attempts: Overall psoriasis: No. events: 63 Incidence rate per 10,000 person-years: 1.43 (95% CI, 1.11-1.83)	Suicide attempts: Overall controls: No. events: 217 Incidence rate per 10,000 person-years: 1.00 (95% CI, 0.88-1.15)	Suicide attempts: OR, 1.44 (95% CI, 1.09-1.91)* Incidence rate ratios (IRR) Overall psoriasis: IRR, 1.42 (95% CI, 1.08-1.88)
			Suicide attempts: Mild psoriasis: No. events: 42 Incidence rate per 10,000 person-years: 1.17 (95% CI, 0.87-1.59)	Suicide attempts: Mild psoriasis-matched general population controls: No. events: 172 Incidence rate per 10,000 person-years: 0.97 (95% CI, 0.84-1.13)	Mild psoriasis: IRR, 1.20 (95% CI, 0.86-1.69)
			Suicide attempts: Severe psoriasis: No. events: 21 Incidence rate per 10,000 person-years: 2.52 (95% CI, 1.65-3.88)	Suicide attempts: Severe psoriasis-matched general population controls: No. events: 45 Incidence rate per 10,000 person-years: 1.14 (95% CI, 0.85-1.52)	Severe psoriasis: IRR, 2.23 (95% CI, 1.33-3.74)
			Completed suicides: Overall psoriasis: No. events: Physician-reported: 107 Police-confirmed: 90 Incidence rate per 10,000 person-years: Physician reported: 2.42 (95% CI, 2.00-2.93)	Completed suicides: Overall controls: No. events: Physician-reported: 467 Police-confirmed: 355 Incidence rate per 10,000 person-years: Physician reported: 2.16 (95% CI, 1.97-2.36)	Completed suicides: OR, 1.19 (95% CI, 1.02-1.39)* IRR Physician-reported: Overall psoriasis: IRR, 1.12 (95% CI, 0.91-1.39)

			Police-confirmed: 2.03 (95% CI, 1.66-2.50)	Police-confirmed: 1.64 (95% CI, 1.48-1.82)	Mild psoriasis: IRR, 1.17 (95% CI, 0.92-1.46) Severe psoriasis: IRR, 0.92 (95% CI, 0.53-1.59)
			Completed suicides: Mild psoriasis: No. events: Physician-reported: 92 Police-confirmed: 76 Incidence rate per 10,000 person-years: Physician reported: 2.57 (95% CI, 2.09-3.15) Police-confirmed: 2.12 (95% CI, 1.69-2.65)	Completed suicides: Mild psoriasis-matched general population controls: No. events: Physician-reported: 389 Police-confirmed: 299 Incidence rate per 10,000 person-years: Physician reported: 2.20 (95% CI, 1.99-2.43) Police-confirmed: 1.69 (95% CI, 1.51-1.90)	Police-confirmed: Overall psoriasis: IRR, 1.24 (95% CI, 0.98-1.66) Mild psoriasis: IRR, 1.25 (95% CI, 0.97-1.61) Severe psoriasis: IRR, 1.19 (95% CI, 0.66-2.14)
			Completed suicides: Severe psoriasis: No. events: Physician-reported: 15 Police-confirmed: 14 Incidence rate per 10,000 person-years: Physician reported: 1.80 (95% CI, 1.09-2.99) Police-confirmed: 1.68 (95% CI, 1.00-2.84)	Completed suicides: Severe psoriasis-matched general population controls: No. events: Physician-reported: 78 Police-confirmed: 56 Incidence rate per 10,000 person-years: Physician reported: 1.97 (95% CI, 1.58-2.46) Police-confirmed: 1.41 (95% CI, 1.09-1.84)	
Pompili et al[30] (2016)	HAM-D and HAM-A	Ideation and attempt	Suicidal ideation: Current: 14/91 (15.4%) Past: 34/91 (37.4%) Suicide attempts: 6/91 (6.6%)	Suicidal ideation: Current: 8/66 (12.1%) Past: 11/66 (16.7%) Suicide attempts: 0/66 (0%)	Suicidal ideation: OR, 2.76 (95% CI, 1.41-5.41)* Suicide attempts: OR, 10.11 (95% CI, 0.55-184.27)*
Lamb et al[31] (2016)	PHQ-9	Ideation	21/607 (3.5%)	NR	NR

Continued

Table II. Cont'd

Study	Outcome ascertainment	Aspects of suicidality assessed	Suicidality in psoriasis, n (%)	Suicidality in controls, n (%)	Psoriasis and suicidality: measure of effect
Wu et al[32] (2017)	PHQ-9, ICD-9 and ICD-10 codes	Ideation, attempt, and completed	Suicidal ideation: Overall psoriasis: No. events: 55 Incidence rate per 1000 person-years: 0.33 Suicide attempts: Overall psoriasis: No. events: 155 Incidence rate per 1000 person-years: 1.31 Any suicidality: Overall psoriasis: No. events: 381 Incidence rate per 1000 person-years: 3.22 Mild psoriasis: No. events: 333 Incidence rate per 1000 person-years: 3.19 Severe psoriasis: No. events: 48 Incidence rate per 1000 person-years: 3.52	Suicidal ideation: General population: No. events: 167 Incidence rate per 1000 person-years: 0.27 Suicide attempts: General population: No. events: 526 Incidence rate per 1000 person-years: 1.15 Any suicidality: General population: No. events: 1199 Incidence rate per 1000 person-years: 2.63	Suicidal ideation: OR, 1.42 (95% CI, 1.05-1.93)* Adjusted IRR, 1.19 (95% CI, 0.87-1.63) Suicide attempts: OR, 1.27 (95% CI, 1.06-1.52)* Adjusted IRR, 1.01 (95% CI, 0.85-1.20) Any suicidality: Adjusted IRRs Overall psoriasis: IRR, 1.10 (95% CI, 0.98-1.24) Mild psoriasis: IRR, 1.10 (95% CI, 0.97-1.25) Severe psoriasis: IRR, 1.09 (95% CI, 0.82-1.44)

AOR, Adjusted odds ratio; *BDI*, Beck Depression Inventory; *CDR*, Causes of Death Register; *CI*, confidence interval; *CRSD*, Carroll Rating Scale for Depression; *GHQ-H*, General Health Questionnaire section "H"; *HAM-A*, Hamilton Anxiety Scale; *HAM-D*, Hamilton Depression Rating Scale; *HR*, hazard ratio; *ICD*, International Classification of Diseases; *IDF*, Israeli Defense Forces; *IRR*, incidence rate ratio; *N/A*, not available; *NR*, not reported; *OR*, odds ratio; *PHQ*, Patient Health Questionnaire; *PSOLAR*, Psoriasis Longitudinal Assessment and Registry.

*In studies that did not report odds ratios, unadjusted odds ratios were calculated using published results.

†Measure of effect unable to be determined due to the prospective cohort nature of the study.

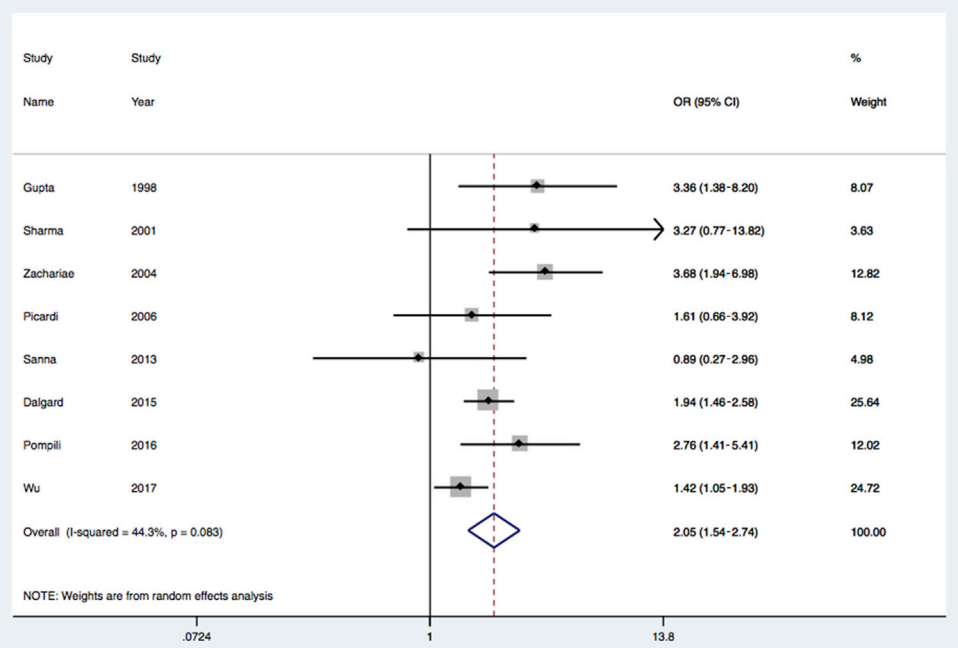


Fig 2. Prevalence of suicidal ideation in patients with psoriasis. This is a forest plot examining observational studies. The diamond represents the exact estimate from the study. The width of the line extending from each diamond represents the 95% confidence interval (CI). *OR*, Odds ratio.

between patients with psoriasis and controls without psoriasis.[15]

Distinct from suicidal ideation, suicidal *behaviors* or *acts* typically refer to suicide attempts and completed suicides. Eleven out of 18 studies examined suicide attempts and completed suicides in patients with psoriasis and cohorts without psoriasis. Pompili et al reported that 6.6% of the patients with psoriasis had attempted suicide at some point during their life versus 0% of the controls without psoriasis but with other skin conditions.[30] One study examined occurrence of new suicide attempts and completed suicides per year among 68,511 patients with psoriasis. The incidence rates of suicide attempts in patients with psoriasis and matched controls were 1.43 and 1.00 per 10,000 person-years, respectively. The incidence rates of completed suicides in patients with psoriasis and matched controls were 2.03 and 1.64 per 10,000 person-years, respectively. Two additional studies examined completed suicide and reported higher prevalence or incidence of suicide completion in patients with psoriasis than in cohorts without psoriasis.[18,27]

After synthesizing data on suicidal behavior from 7 studies, we found that compared with controls without psoriasis, patients with psoriasis are more likely to attempt and complete suicides (pooled OR, 1.26; 95% CI, 1.13-1.40) (Fig 3). A funnel plot of these studies suggested limited publication bias, which was confirmed by the Egger regression test ($P = .101$) (Supplemental Fig 2; available at http://www.jaad.org). The I^2 value of 0 suggests limited heterogeneity. On the basis of our sensitivity analysis examining suicide attempts and complete suicides separately, patients with psoriasis have a higher likelihood of suicide attempts (pooled OR, 1.32; 95% CI,1.14-1.54) and completed suicide (pooled OR, 1.20; 95% CI, 1.04-1.39) than patients without psoriasis.

Three studies stratified patients with psoriasis by age when examining suicidality.[16,18,24] The studies reported that younger patients were more likely to experience suicidality than older patients. This was evidenced by a large study of 916,948 participants, which found that compared with controls without psoriasis, patients with psoriasis who were in their 20s had a higher risk of suicidality (HR, 1.83; 95% CI, 1.64-2.05) than patients with psoriasis aged 60 years or older (HR, 1.04; 95% CI, 0.91-1.20).[24]

In 5 studies, the authors stratified patients with psoriasis by disease severity.[16,24,27,29,32] Three studies found that patients with severe psoriasis are more likely to experience suicidal ideation and suicide attempts than controls without psoriasis.[16,19,24] Specifically, Rapp et al reported that patients with psoriasis with suicidal ideation had higher disease severity scores than patients with psoriasis without suicidal ideation (Self-Administered Psoriasis Area and Severity Index score

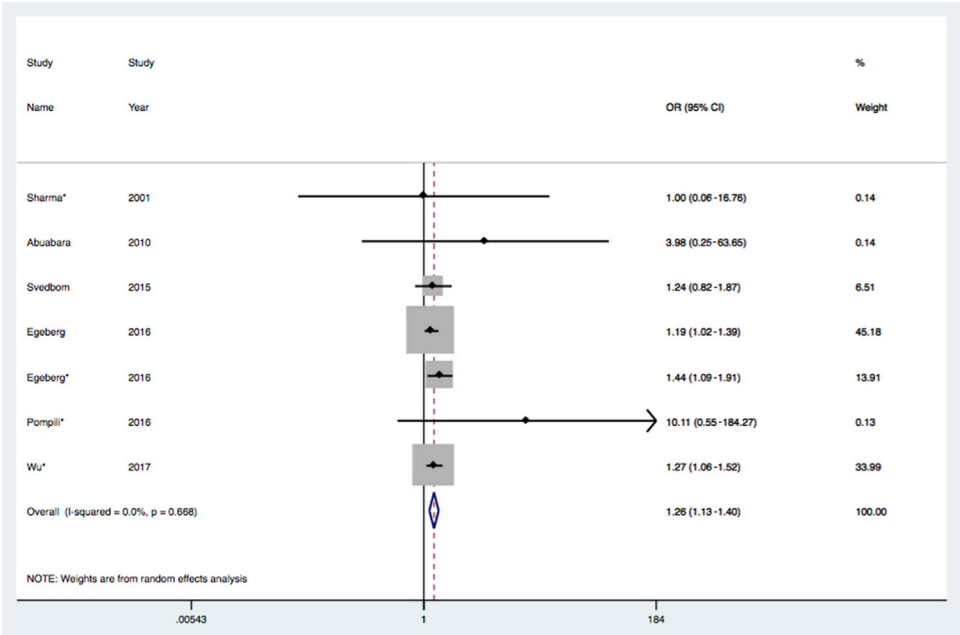


Fig 3. Prevalence of suicidal behavior (suicide attempts and completed) in patients with psoriasis. This is a forest plot examining observational studies. The diamond represents the exact estimate from the study. The width of the line extending from each diamond represents the 95% confidence interval (CI). Asterisk indicates study examining suicide attempts. *OR*, Odds ratio.

of 11.47 vs. 6.23, respectively).[16] One study examining suicide attempts and suicide completion found that compared with controls, patients with mild psoriasis disease had a lower rate of suicide attempts (1.17 per 10,000 person-years) than patients with severe psoriasis (2.52 per 10,000 person-years). A dose-dependent relationship was not seen for suicide completion.[29]

DISCUSSION

Population-based studies have posited a relationship between psoriasis and psychiatric comorbidities.[10,33-35] This systematic review has examined the literature to assess data from 18 studies with a total of 1,767,583 study participants, of whom 330,207 were patients with psoriasis.

This study showed that patients with psoriasis have increased odds of all aspects of suicidality (ideation, attempts, and completion) compared with the general population. This finding was most pronounced for suicidal ideation. Patients with psoriasis are twice as likely to contemplate suicide a people without psoriasis. Additionally, patients with psoriasis had a higher likelihood of suicide attempts and completed suicides. Patients with psoriasis had a 32% higher likelihood of attempting suicide than those without psoriasis. They also had a 20% higher likelihood of completed suicide than people without psoriasis.

Patients with more severe psoriasis were more likely to contemplate and attempt suicides. Greater severity of psoriasis has been associated with greater social stigmatization, substantially impaired quality of life, and higher depression rates, all of which can contribute to greater suicidal ideation and behavior. Thus, controlling psoriasis will not only have a significant impact on morbidity but can also affect suicide-related morality. This finding also highlights the importance of establishing effective treatment goals to monitor and reduce the severity of psoriasis disease.[36]

Another key finding from this investigation is that compared with patients without psoriasis, younger patients with psoriasis are especially at risk for suicidal ideation, suicidal attempts, and completed suicides. Although little is known regarding the reason for this, international studies involving the general population have shown that when the data are adjusted for socioeconomic status and degree of hardship, younger patients are more likely to commit suicide than older patients.[37,38] It is possible that many younger patients with psoriasis have not yet developed effective disease-coping mechanisms or may be more prone to impulsive behavior. Therefore, interventions aimed at helping young adults cope with psoriasis and its mental comorbidities are paramount.

Several possible biologic mechanisms have been proposed to explain the epidemiologic association between psoriasis and psychiatric comorbidities.[39] Multiple studies have reported an association between the proinflammatory cytokines interleukin 6 (IL-6), IL-1, IL-17, tumor necrosis factor-α, and depression.[40-46] Studies examining this link have identified higher blood and cerebrospinal fluid levels of proinflammatory cytokines in patients with depression who are otherwise medically healthy and without inflammatory skin disease. It is believed that elevated cytokine levels may affect serotonin metabolism.

To increase understanding of the effect of proinflammatory cytokines on suicidal behavior, studies have examined levels of cytokines in the cerebrospinal fluid and brain tissue of people who attempt or complete suicide.[47,48] One study reported higher levels of IL-6 in the cerebrospinal fluid of those who attempted suicide than in healthy controls who did not attempt suicide.[48] An elevated inflammatory state, as seen in psoriasis, may affect the development and progression of depressive symptoms and suicidal behavior; however, the exact biologic mediators involved are under active investigation.

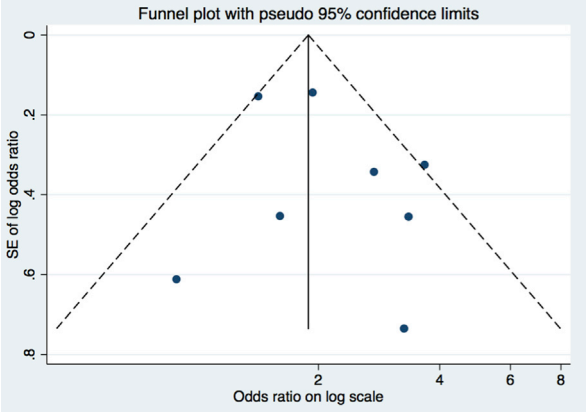
This systematic review and meta-analysis needs to be interpreted in the context of the available primary studies. As this was a meta-analysis of study-level data, there were possible sources of heterogeneity and bias among the included studies. There was variability in the methodology of assessing and reporting suicidality. Additionally, a minority of the studies examined prevalence of completed suicides in patients with psoriasis versus in controls. There were multiple uncontrolled studies in this systematic review that were also limited in terms of study quality when assessed by using the Newcastle-Ottawa scale. To account for heterogeneity among the studies, we utilized random effects analysis to calculate pooled ORs.

In conclusion, this systematic review and meta-analysis showed that patients with psoriasis have a significantly higher likelihood of experiencing suicidal ideation, suicide attempts, and completed suicides. Patients with psoriasis with greater disease severity and younger patients are especially at risk for suicidality. These findings suggest a need for providers to inquire about suicidality in patients with psoriasis. Caring for patients with psoriasis and their mental health needs may require a concerted effort among dermatologists and other health care providers.

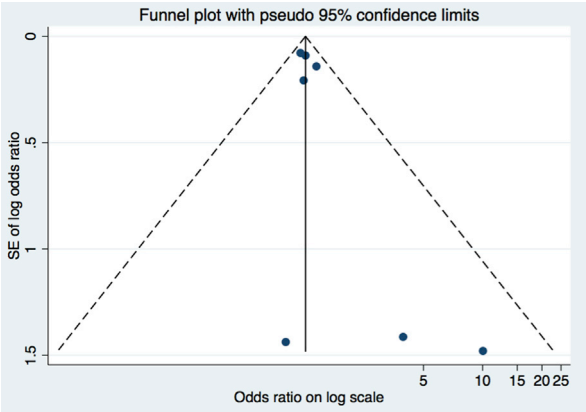
REFERENCES

1. Rachakonda TD, Schupp CW, Armstrong AW. Psoriasis prevalence among adults in the United States. *J Am Acad Dermatol*. 2014;70(3):512-516.
2. Parisi R, Symmons DP, Griffiths CE, et al. Global epidemiology of psoriasis: a systematic review of incidence and prevalence. *J Invest Dermatol*. 2013;133(2):377-385.
3. Stern RS, Nijsten T, Feldman SR, Margolis DJ, Rolstad T. Psoriasis is common, carries a substantial burden even when not extensive, and is associated with widespread treatment dissatisfaction. *J Invest Dermatol Symp Proc*. 2004;9(2):136-139.
4. Singh S, Young P, Armstrong AW. Relationship between psoriasis and metabolic syndrome: a systematic review. *G Ital Dermatol Venereol*. 2016;151(6):663-677.
5. Mehta V, Malhotra SK. Psychiatric evaluation of patients with psoriasis vulgaris and chronic urticaria. *German J Psychiatry*. 2007;10(4):104-110.
6. Bewley A, Burrage DM, Ersser SJ, Hansen M, Ward C. Identifying individual psychosocial and adherence support needs in patients with psoriasis: a multinational two-stage qualitative and quantitative study. *J Eur Acad Dermatol Venereol*. 2014;28(6):763-770.
7. Ring L, Kettis-Lindblad A, Kjellgren KI, Kindell Y, Maroti M, Serup J. Living with skin diseases and topical treatment: patients' and providers' perspectives and priorities. *J Dermatolog Treat*. 2007;18(4):209-218.
8. Dowlatshahi EA, Wakkee M, Arends LR, Nijsten T. The prevalence and odds of depressive symptoms and clinical depression in psoriasis patients: a systematic review and meta-analysis. *J Invest Dermatol*. 2014;134(6):1542-1551.
9. Fleming P, Bai JW, Pratt M, Sibbald C, Lynde C, Gulliver WP. The prevalence of anxiety in patients with psoriasis: a systematic review of observational studies and clinical trials. *J Invest Dermatol*. 2016;136(9):S162.
10. Koo J. Population-based epidemiologic study of psoriasis with emphasis on quality of life assessment. *Dermatol Clin*. 1996;14(3):485-496.
11. Stang A. Critical evaluation of the Newcastle-Ottawa scale for the assessment of the quality of nonrandomized studies in meta-analyses. *Eur J Epidemiol*. 2010;25(9):603-605.
12. Herzog R, Álverez-Pasquin MJ, Diaz C, Del Barrio JL, Estada JM, Gil Á. Are healthcare workers' intentions to vaccinate related to their knowledge, beliefs and attitudes? A systematic review. *BMC Public Health*. 2013;13:154.
13. Szumilas M. Explaining odds ratios. *J Can Acad Child Adolesc Psychiatry*. 2010;19(3):227-229.
14. Brockwell SE, Gordon IR. A comparison of statistical methods for meta-analysis. *Stat Med*. 2001;20(6):825-840.
15. Gupta MA, Schork NJ, Gupta AK, Kirkby S, Ellis CN. Suicidal ideation in psoriasis. *Int J Dermatol*. 1993;32(3):188-190.
16. Rapp SR, Exum ML, Reboussin DM, Feldman SR, Fleischer A, Clark A. The physical, psychological and social impact of psoriasis. *J Health Psychol*. 1997;2(4):525-537.
17. Gupta MA, Gupta AK. Depression and suicidal ideation in dermatology patients with acne, alopecia areata, atopic dermatitis and psoriasis. *Br J Dermatol*. 1998;139(5):846-850.
18. Krueger G, Koo J, Lebwohl M, Menter A, Stern RS, Rolstad T. The impact of psoriasis on quality of life—Results of a 1998 National Psoriasis Foundation Patient-Membership Survey. *Arch Dermatol*. 2001;137(3):280-284.
19. Sharma N, Koranne RV, Singh RK. Psychiatric morbidity in psoriasis and vitiligo: a comparative study. *J Dermatol*. 2001;28(8):419-423.
20. Zachariae R, Zachariae C, Ibsen HH, Mortensen JT, Wulf HC. Psychological symptoms and quality of life of dermatology

outpatients and hospitalized dermatology patients. *Acta Derm Venereol*. 2004;84(3):205-212.
21. Friedman T, Wohl Y, Knobler HY, et al. Increased use of mental health services related to isotretinoin treatment: a 5-year analysis. *Eur Neuropsychopharmacol*. 2006;16(6):413-416.
22. Picardi A, Mazzotti E, Pasquini P. Prevalence and correlates of suicidal ideation among patients with skin disease. *J Am Acad Dermatol*. 2006;54(3):420-426.
23. Abuabara K, Azfar RS, Shin DB, Neimann AL, Troxel AB, Gelfand JM. Cause-specific mortality in patients with severe psoriasis: a population-based cohort study in the U.K. *Br J Dermatol*. 2010;163(3):586-592.
24. Kurd SK, Troxel AB, Crits-Christoph P, Gelfand JM. The risk of depression, anxiety, and suicidality in patients with psoriasis: a population-based cohort study. *Arch Dermatol*. 2010;146(8):891-895.
25. Sanna L, Stuart AL, Pasco JA, et al. Suicidal ideation and physical illness: does the link lie with depression? *J Affect Disord*. 2014;152-154:422-426.
26. Kimball AB, Leonardi C, Stahle M, et al. Demography, baseline disease characteristics and treatment history of patients with psoriasis enrolled in a multicentre, prospective, disease-based registry (PSOLAR). *Br J Dermatol*. 2014;171(1):137-147.
27. Svedbom A, Dalén J, Mamolo C, et al. Increased cause-specific mortality in patients with mild and severe psoriasis: a population-based Swedish register study. *Acta Derm Venereol*. 2015;95(7):809-815.
28. Dalgard FJ, Gieler U, Tomas-Aragones L, et al. The psychological burden of skin diseases: a cross-sectional multicenter study among dermatological out-patients in 13 European countries. *J Invest Dermatol*. 2015;135(4):984-991.
29. Egeberg A, Hansen PR, Gislason GH, Skov L, Malbris L. Risk of self-harm and nonfatal suicide attempts, and completed suicide in patients with psoriasis: a population-based cohort study. *Br J Dermatol*. 2016;175(3):493-500.
30. Pompili M, Innamorati M, Trovarelli S, et al. Suicide risk and psychiatric comorbidity in patients with psoriasis. *J Int Med Res*. 2016;44(suppl 1):61-66.
31. Lamb RC, Matcham F, Turner MA, et al. Screening for anxiety and depression in people with psoriasis: a cross sectional study in a tertiary referral setting. *Br J Dermatol*. 2016;176:1028-1034.
32. Wu JJ, Penfold RB, Primatesta P, et al. The risk of depression, suicidal ideation and suicide attempt in patients with psoriasis, psoriatic arthritis or ankylosing spondylitis. *J Eur Acad Dermatol Venereol*. 2017;175:493.
33. Schmitt JM, Ford DE. Role of depression in quality of life for patients with psoriasis. *Dermatology*. 2007;215(1):17-27.
34. Fortune DG, Richards HL, Kirby B, Bowcock S, Main CJ, Griffiths CE. A cognitive-behavioural symptom management programme as an adjunct in psoriasis therapy. *Br J Dermatol*. 2002;146(3):458-465.
35. Golpour M, Hosseini SH, Khademloo M, et al. Depression and anxiety disorders among patients with psoriasis: a hospital-based case-control study. *Dermatol Res Pract*. 2012;2012:381905.
36. Armstrong AW, Siegel MP, Bagel J, et al. From the Medical Board of the National Psoriasis Foundation: treatment targets for plaque psoriasis. *J Am Acad Dermatol*. 2017;76(2):290-298.
37. Shah A. Suicide rates: age-associated trends and their correlates. *J Inj Violence Res*. 2012;4(2):79-86.
38. Stockard J, O'Brien RM. Cohort variations and changes in age-specific suicide rates over time: explaining variations in youth suicide. *Social Forces*. 2002;81(2):605-642.
39. Pandey GN. Biological basis of suicide and suicidal behavior. *Bipolar Disord*. 2013;15(5):524-541.
40. Howren MB, Lamkin DM, Suls J. Associations of depression with C-reactive protein, IL-1, and IL-6: a meta-analysis. *Psychosom Med*. 2009;71(2):171-186.
41. Dowlati Y, Herrmann N, Swardfager W, et al. A meta-analysis of cytokines in major depression. *Biol Psychiatry*. 2010;67(5):446-457.
42. Raison CL, Capuron L, Miller AH. Cytokines sing the blues: inflammation and the pathogenesis of depression. *Trends Immunol*. 2006;27(1):24-31.
43. Davami MH, Baharlou R, Vasmehjani AA, et al. Elevated IL-17 and TGF-beta serum levels: a positive correlation between T-helper 17 cell-related pro-inflammatory responses with major depressive disorder. *Basic Clin Neurosci*. 2016;7(2):137-142.
44. Martin DA, Towne JE, Kricorian G, et al. The emerging role of IL-17 in the pathogenesis of psoriasis: preclinical and clinical findings. *J Invest Dermatol*. 2013;133(1):17-26.
45. Wang J, Wang YMC, Ahn HY. Biological products for the treatment of psoriasis: therapeutic targets, pharmacodynamics and disease-drug-drug interaction implications. *AAPS J*. 2014;16(5):938-947.
46. Gimeno D, Kivimäki M, Brunner EJ, et al. Associations of C-reactive protein and interleukin-6 with cognitive symptoms of depression: 12-year follow-up of the Whitehall II study. *Psychol Med*. 2009;39(3):413-423.
47. Hung YY, Kang HY, Huang KW, Huang TL. Association between toll-like receptors expression and major depressive disorder. *Psychiatry Res*. 2014;220(1-2):283-286.
48. Lindqvist D, Janelidze S, Hagell P, et al. Interleukin-6 is elevated in the cerebrospinal fluid of suicide attempters and related to symptom severity. *Biol Psychiatry*. 2009;66(3):287-292.



Supplemental Fig 1. Funnel plot of the studies examining suicidal ideation in patients with psoriasis. *SE*, Standard error.



Supplemental Fig 2. Funnel plot of the studies examining suicidal behavior (suicide attempts and completed suicides) in patients with psoriasis. *SE*, Standard error.